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Gavin L. Moir

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Three Different Methods of Calculating Vertical Jump Height from Force Platform Data in Men and Women

Gavin L. Moir
Exercise Science Department
East Stroudsburg University of Pennsylvania
East Stroudsburg, Pennsylvania

The purpose of the present study was to investigate the effects of different methods to calculate vertical jump height in men and women. Fifty men and 50 women performed three countermovement vertical jumps for maximal height on a force platform, the highest of which was used in the statistical analyses. The peak displacement attained by the center of mass (COM) during flight was obtained from three different calculations: (1) using the time in the air (TIA), (2) using the vertical velocity of the COM at take-off (TOV), and (3) adding the positive vertical displacement of the COM prior to take-off to the height calculated using TOV (TOV+s). With all calculations, men produced significantly greater jump heights than women ($p < 0.05$). TIA produced significantly greater jump heights than TOV in men and women, while TOV+s produced significantly greater jump heights than both TIA and TOV in men and women ($p < 0.05$). Despite these differences, the methods produced consistent results for both men and women. All calculation methods have logical validity, depending upon the definition of jump height used. Therefore, the method used to calculate jump height should be determined by the equipment available to the practitioner while giving consideration to the sources of error inherent in each method. Based upon the present findings, when using a force platform to calculate vertical jump height, practitioners are encouraged to use the TOV method.

Key words: Vertical jump, displacement, take-off velocity, uniform acceleration

Correspondence should be sent to Gavin L. Moir, Ph.D., Exercise Science Department, East Stroudsburg University of Pennsylvania, East Stroudsburg, PA, 18301-2999. E-mail: gmoir@po-box. esu.edu

Vertical jumps form an important test to assess the explosive strength of the leg musculature of athletes (Tidow, 1990; Young, 1995), with one of the most common variables calculated being jump height. There are a number of methods used to calculate the jump height achieved during the test, depending upon the equipment used. Field tests, such as the standing jump board and Vertec, provide simple tools to assess jump height. However, because of the arm movements during the performances, the validity of these field tests as a measure of explosive leg strength has been questioned (Young, MacDonald, & Flowers, 2001). From the time in the air (TIA), the vertical displacement of the center of mass (COM) can be calculated using an equation of uniform acceleration. This is the principle behind the use of contact mats that do not require arm movements during the jumps. Contact mats have been shown to be a reliable field test for assessing jump height in both men and women (Moir, Shastri, & Connaboy, 2008). Despite the appeal of this method for calculating jump height, an assumption with the equation of uniform acceleration used is that the position of the COM is the same at the beginning of the jump (take-off) as it is at the end of the jump (landing). This assumption gives rise to jump heights being artificially increased if the subject alters the landing posture, and, therefore, the validity of calculating jump height from the time in the air has been questioned (see Hatze, 1998).

Through the use of a force platform, the vertical velocity of the COM at take-off (TOV) can be calculated by integrating the vertical force trace. Again, an equation of uniform acceleration can be used to calculate the jump height, and this method has been shown to be reliable in men and women (Moir, 2008). Although this method is not affected by the differences in the position of the COM at take-off and landing, it fails to take into account the change in the vertical position of COM prior to take-off caused by joint extension. Previously, Aragón-Vargas (2000) noted that the vertical displacement of the COM prior to take-off can be calculated using video-based motion analysis systems. However, the vertical displacement of the COM prior to take-off can also be calculated through the double integration of the vertical force trace collected from a force platform. This positive displacement can then be added to that calculated from TOV to provide a measure of the total vertical displacement of the COM during a vertical jump (method TOV+s). This method of calculating jump height has also been shown to be reliable in men and women (Moir, 2008).

The use of these different methods for calculating jump height will depend upon the equipment available to the practitioner, as well as the definition of jump height achieved during a vertical jump. The latter is concerned with what can be considered the logical validity of the test (Baumgartner, Jackson, Mahar, & Rowe, 2007). Some researchers have defined jump height as the vertical displacement achieved by the COM from take-off to the vertex of the flight trajectory (Bosco, Luhtanen, & Komi, 1983). From this definition, jump height can be calculated from the time of flight, and, therefore, the use of a simple contact mat or a force platform would allow the calculation of jump height. Others, however, suggest that jump height depends on both take-off velocity and the COM position at take-off, and that jump height will be affected by both of these parameters (Bobbert & van Ingen Schenau, 1988). Using this approach, only the addition of the peak vertical displacement achieved by the COM during flight and the vertical displacement of the COM prior to take-off will suffice when calculating jump height. Therefore, jump height could only be accurately measured using more sophisticated equipment such as a motion analysis system or a force platform.

Aragón-Vargas (2000), using a force platform, reported that the heights calculated from TIA were significantly greater than those calculated using TOV in the jumps of physically active men. It was also reported that the TOV+s method, which used a motion analysis system to calculate the vertical displacement of the COM prior to take-off and a force platform to calculate TOV, produced jump heights greater than those calculated using both TIA and TOV. Despite the theoretical underpinnings of the TOV+s method, Aragón-Vargas (2000) noted that the accuracy of this calculation was poor when using the vertical displacement of the COM during the jump gathered from a motion analysis system as the criterion measurement. It was suggested that some of the error in the TOV+s method may have arisen from the inability to synchronize the force platform and video signals. However, as noted previously, through the double integration of the vertical force trace data from a force platform, the synchronization issue is eliminated.

The purpose of the present study was to investigate the differences in jump heights gathered from a force platform using three different methods of calculation in men and women. The degree of consistency between the different calculation methods was calculated, while regression analyses were also performed using TOV+s as the criterion measure based upon the definition of jump height provided by Bobbert & van Ingen Schenau (1988).

METHOD

Data Collection

Fifty recreationally active men (mean age = 21.7 ± 2.19 years; mean height = 1.78 ± 0.07 m; mean body mass = 82.6 ± 13.2 kg) and 50 recreationally active women (mean age = 20.8 ± 1.90 years; mean height = 1.66 ± 0.06 m; mean body mass = 66.2 ± 12.0 kg) volunteered to participate in this study, which was approved by the Institutional Review Board for the Protection of Human Subjects of East Stroudsburg University. Following a 5-min warm-up consisting of dynamic exercises, each subject performed three countermovement vertical jumps for maximal height on a force platform (Kistler, type 9286AA, Winterthur, Switzerland). Data were sampled at 1,202 Hz, as a frequency of this magnitude has been shown to reduce the error associated with vertical jump height calculated from a force platform (Street, McMillan, Board, Rasmussen, & Heneghan, 2001). The analogue signal from the force platform was then converted to a digital signal using Bioware software (Kistler, Amherst, NY). The data were not filtered, as this has been identified as a potential source of error when calculating jump height (Street et al., 2001). Each subject placed their hands around their neck during each jump in order to remove the influence of the arms, and 4 min of rest were provided between jumps. Prior to each testing session, the force platform was calibrated using loads of known weight.

Data Analysis

Five seconds of data were collected during each jump. The subject's body weight was calculated by averaging the vertical force trace over the first 1 sec of data collection (1,202 data points) when the subject was stationary on the platform (Figure 1). The start of the movement was identified by calculating the peak residual from the vertical force trace during the 1 sec of quiet stance (peak difference between the vertical force trace and the subject's body weight). Specifically, the start of the movement was identified as the first time instant when the vertical force trace was greater (less) than the addition (subtraction) of the peak residual during quiet stance and the subject's body weight, depending upon whether the initial movement by the subject caused an increase or decrease of vertical force.

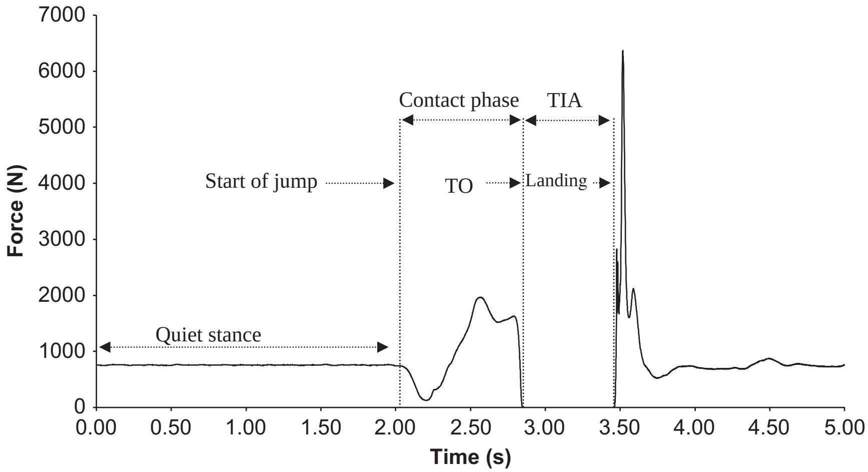

FIGURE 1 The vertical force trace from a subject performing a countermovement vertical jump.
Note. TIA = time in air; TO = take-off.

From the start of the movement, the vertical force trace was then taken back until a value within one Newton of the subject's body weight was identified, and this time was used as the start of the jump. Take-off was identified by calculating the peak residual across a 0.3-sec period (361 data points) during the flight phase of the jump with the force platform unloaded (peak difference between the vertical force trace and 0 N). This period was chosen, as all subjects were able to produce a flight phase greater than 0.3 sec. The peak residual during flight was used to identify both take-off (when the vertical force trace < peak residual during flight) and contact after flight (when the vertical force trace was > peak residual during flight).

Three methods of calculating jump height were used: (1) time in the air (TIA), (2) vertical velocity of the COM at take-off (TOV), and (3) adding the vertical displacement of the COM prior to take-off to the height calculated using TOV (TOV+s). Time in the air was identified as the period between take-off and contact after flight (Figure 1). The time was then used in the following equation of uniform acceleration:

$$\text{TIA jump height} = {}^{1}\!/_{2}\, g(t/2)^2, \tag{1}$$

where $g = 9.81\ \text{m} \cdot \text{sec}^{-2}$, t = time in air.

In order to calculate the vertical velocity of the COM, the vertical force trace was integrated using the trapezoid rule once the trace had been normalized to body mass. Integration began at the start of the jump and ended at take-off. The duration of integration and the method of integration (e.g., trapezoid versus Simpson's approximation) have been shown to contribute very little to the total error in take-off velocity during vertical jumps performed on a force platform (Street et al., 2001). The vertical velocity of the COM at take-off (Figure 2) was then used in the following equation of uniform acceleration to calculate jump height:

$$\text{TOV jump height} = \text{TOV}^2 / 2g, \tag{2}$$

where TOV = vertical velocity of the COM at take-off, $g = 9.81\ \text{m} \cdot \text{sec}^{-2}$.

To calculate vertical displacement of the COM during the time when the subject was in contact with the force platform, the vertical force trace was double integrated. The positive vertical displacement of the COM prior to take-off (Net COM; see Figure 2) was then added to the jump height calculated from TOV (Equation 2). This allowed the calculation of total positive vertical displacement of the COM from the subject's starting position (TOV+s).

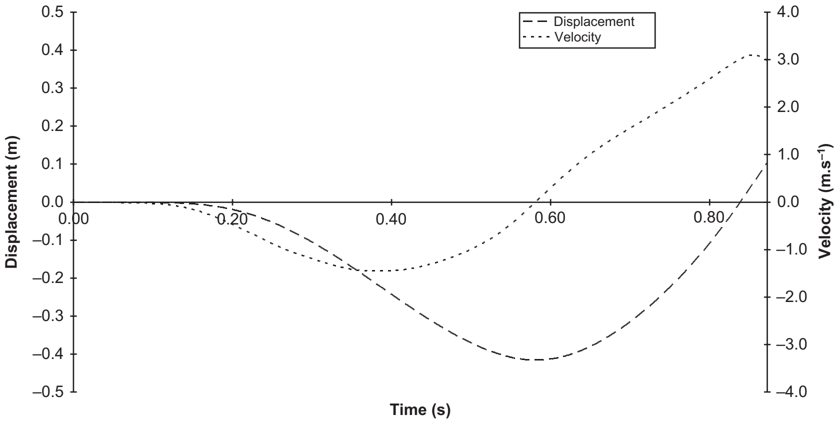

FIGURE 2 The vertical displacement and velocity of the center of mass of a subject during the contact phase of a countermovement vertical jump.
Note. The vertical displacement of the center of mass rises above that during quiet stance prior to take-off. Also, the vertical velocity of the center of mass falls just before take-off.

From the three trials performed by each subject, the trial yielding the greatest height as calculated using the TOV+s method was used in the analysis for all calculation methods. In this manner, TOV+s was used as the criterion measurement for jump height in this study based upon the definition of jump height provided by Bobbert & van Ingen Schenau (1988).

Statistical Analysis

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS for Windows, version 15.0, SPSS Inc., Chicago, IL). Measures of central tendency and spread of the data were represented as means and standard deviations. Coefficients of variation (CV) for each calculation method were also provided as the ratio of the standard deviation to the mean value expressed as a percentage (100 *SD*/Mean). Differences between the three methods for calculating jump height and the differences between the sexes were identified using analyses of variance (ANOVA) with repeated measures on one factor (jump height calculation) and one between subjects factor (gender). Pairwise comparisons with Bonferroni corrections were conducted to identify where the differences in jump height calculations occurred. The difference in Net COM between men and women was assessed using an independent *t*-test. The degree of consistency between the calculation methods was assessed using intraclass correlation coefficients (ICC) calculated from a two-way mixed method consistency model for single measures. Regression coefficients and prediction errors were calculated using simple linear regression models, with the TOV+s method as the dependent variable as it is the criterion measure in the present study. The regression analyses provide a measure of predictive validity for the TIA and TOV calculation methods. Alpha was set at $p \leq 0.05$ for all analyses, and the 95% confidence intervals (95% CI) were calculated where appropriate.

RESULTS

Tables 1 and 2 show the mean jump heights calculated by using TIA, TOV, and TOV+s for both men and women, respectively, as well as the Net COM.

There was a significant main effect for gender ($F(1, 98) = 221.526$, $p < 0.001$), with the men producing greater jump heights than the women. The mean difference in jump heights between the genders was 0.149 m, (95% CI: 0.129 – 0.169 m). A significant main effect was reported for jump height calculation method ($F(1.729, 169.470) = 202.230$, $p < 0.001$). Jump heights calculated using TIA

TABLE 1
Descriptive Statistics for the Jump Heights for Men ($N = 50$)

	TIA	*TOV*	*TOV+s*	*Net COM*
Mean (m)	0.358	0.348	0.467	0.119
SD (m)	0.059	0.056	0.056	0.021
CV (%)	16.5	16.1	12.0	17.7

Note. TIA = time in air; TOV = take-off velocity; TOV+s = take-off velocity with the positive displacement of the center of mass added; Net COM = the positive displacement of the center of mass prior to take-off; *SD* = standard deviation; CV = coefficient of variation.

TABLE 2
Descriptive Statistics for the Jump Heights for Women ($N = 50$)

	TIA	*TOV*	*TOV+s*	*Net COM*
Mean (m)	0.216	0.207	0.307	0.100
SD (m)	0.046	0.046	0.047	0.018
CV (%)	21.3	22.2	15.3	17.8

Note. TIA = time in air; TOV = take-off velocity; TOV+s = take-off velocity with the positive displacement of the center of mass added; Net COM = the positive displacement of the center of mass prior to take-off; *SD* = standard deviation; CV = coefficient of variation.

were significantly greater than those calculated using TOV (mean difference 0.013 m; 95% CI: 0.010 – 0.017 m). Similarly, jump heights calculated using TOV+s were significantly greater than those calculated using TIA (mean difference 0.094 m; 95% CI: 0.088 – 0.099 m) and TOV (mean difference 0.107 m; 95% CI: 0.103 – 0.111 m). There was a significant gender-calculation method interaction ($F(1.729, 169.470) = 15.410$, $p < 0.001$). Comparisons with Bonferroni corrections showed that men produced significantly higher jumps compared to women regardless of the calculation method used ($p < 0.017$). There was no significant difference in Net COM between the men and women ($p > 0.05$).

Despite the differences in jump height produced by the different calculation methods, a high degree of consistency was demonstrated across the methods in both men (ICC 0.927; 95% CI: 0.887 – 0.955) and women (ICC 0.934; 95% CI: 0.897 – 0.960).

The results of the simple regression models are shown in Tables 3 and 4 for men and women, respectively. For both men and women, the TIA and TOV methods

TABLE 3
Simple Regression Model for Jump Height in Men ($N = 50$)

Model[a]	*R*	R^2	*MSE*	*Error*
TOV+s = 0.162 + 0.854 TIA[b]	0.897	0.805	0.625E-03	0.025
TOV+s = 0.141 + 0.938 TOV[c]	0.929	0.864	0.438E-03	0.021

Note. MSE = mean square error, calculated as the residual sum of squares divided by the degrees of freedom; Error = estimation error, calculated as the square root of MSE; TOV + s = take-off velocity with the positive displacement of the center of mass added; TIA = time in air; TOV = take-off velocity.

[a]All models are significant ($p < 0.05$).

[b]95% confidence interval of slope = 0.732 – 0.976.

[c]95% confidence interval of slope = 0.830 – 1.046.

TABLE 4
Simple Regression Model for Jump Height in Women ($N = 50$)

Model[a]	*R*	R^2	*MSE*	*Error*
TOV+ s = 0.108 + 0.922 TIA[b]	0.905	0.818	0.417E-03	0.020
TOV+ s = 0.109 + 0.956 TOV[c]	0.928	0.861	0.313E-03	0.018

Note. MSE = mean square error, calculated as the residual sum of squares divided by the degrees of freedom; Error = estimation error, calculated as the square root of MSE; TOV + s = take-off velocity with the positive displacement of the center of mass added; TIA = time in air; TOV = take-off velocity.

[a]All models are significant ($p < 0.05$).

[b]95% confidence interval of slope = 0.796 – 1.048.

[c]95% confidence interval of slope = 0.844 – 1.067.

could significantly predict jump height calculated using TOV+s, the criterion measure ($p < 0.05$), explaining more than 80% of the variance in TOV+s jump height. However, in the model using TIA for men, it was found that the slope of the regression line was different from 1.0 (coefficient of slope = 0.854; 95% CI = 0.732 – 0.976). In all other models the slope coefficients were not different from 1.0.

DISCUSSION

The purpose of the present study was to investigate the differences in jump heights gathered from a force platform using three different methods of calculation in men and women. Irrespective of gender, the TOV+s method produced significantly greater jump heights than both the TIA and TOV methods. This is to be expected, as the TOV+s method includes the vertical displacement undergone by the COM prior to take-off. A similar finding was reported by Aragón-Vargas (2000) investigating vertical jumps in physically active men using synchronized force platform and video data. The mean jump height reported in the present study for the TOV+s method (0.467 m) was below the mean value of 0.520 m reported by Aragón-Vargas (2000) despite the mean jump height for the TOV method being very similar between the studies (0.348 m and 0.361 m). While the mean Net COM for the men in the present study was 0.119 m, Aragón-Vargas (2000) reported a mean value of 0.144 m, which may explain some of the difference. However, Aragón-Vargas (2000) noted that problems in attempting to synchronize the video and force platform data in his protocol could lead to overestimations of Net COM to the magnitude of 0.044 m. The present protocol of calculating Net COM from the force platform data overcomes the problems arising from synchronization errors and may provide a more accurate measure of the rise of the COM prior to take-off.

The Net COM for the women in the present study was not significantly different from that of the men despite the significantly greater jump heights achieved by the men. As the TOV+s method of calculating jump height combines the vertical displacement achieved by the COM due to its vertical velocity at take-off and the vertical displacement of the COM prior to take-off achieved by the extension of the joints, then the greater performances achieved by the men can be attributed to the greater vertical velocities of the COM at take-off. It would appear that the vertical displacement of the COM prior to take-off is not as important as take-off velocity in vertical jump performance, confirming previous research findings (Aragón-Vargas & Gross, 1997).

For both men and women in the present study, the TIA method produced significantly greater jump heights than the TOV method, with the magnitude of the differences being similar for men and women (~ 3% and ~ 4%, respectively). Greater jump heights calculated using TIA compared to TOV were also reported by Aragón-Vargas (2000) using force platform data during jumps performed by physically active men. Although both calculation methods use equations of uniform acceleration, the TIA method assumes that the vertical position of the COM upon landing is the same as at take-off, such that the time up to the vertex of flight is equal to the time of descent. This assumption is violated if the jumper changes posture during flight and can be exploited by the subjects attempting to "tuck" their legs slightly during the jump, inflating jump height erroneously. Jump heights calculated from the TOV method are unaffected by this assumption. Some authors have suggested that jump heights for physically active men, measured using a contact mat, are prone to this kind of systematic error (Moir et al., 2008). Interestingly, the same effect has not been reported for physically active women. It is possible that a change in posture during flight accounted for the differences in jump heights found in the present study.

Although Aragón-Vargas (2000) also reported greater jump heights in men using TIA when compared to the TOV method, the increase in jump height in his study as a result of TIA (~ 11%) was much greater than that reported in the present study (~ 3%). One reason to explain this difference may be due to greater changes in take-off and landing postures of the jumpers used. However, there are other explanations pertaining to the equipment used. For example, Aragón-Vargas (2000) used a force platform sampling at 300 Hz. Street et al. (2001) reported that this sampling frequency could cause a 2% underestimation of jump height using the TOV method due to the late selection of the take-off event, which will equate to a lower take-off velocity (see Figure 2). This error is almost eliminated when the sampling frequency is increased to greater than 1080 Hz. This problem is unlikely to affect TIA calculated from the force platform, as both the take-off and landing events are likely to be selected late. Similarly, small errors in the measurement of the subject's body weight from the force platform can produce large errors in jump height calculated from TOV. For example, a 0.25% overestimation in body weight can lead to a 6.5% underestimation in jump height due to the integration process (Street et al., 2001). This error will not affect jump height calculated from TIA. Although Aragón-Vargas (2000) did not fully explain the protocol used, in the present study, body weight was averaged across a 1-sec period of quiet stance in order to reduce this error. Other variables relating to force platforms (e.g., strain-gauge versus piezoelectric force transducers, differences in the damping factors) could also contribute to the greater magnitude in jump height differences using the TIA and TOV methods reported by Aragón-Vargas (2000).

Despite the significant differences in the jump heights, there was a high degree of consistency between the different calculation methods for both men and women as shown in the large ICC values. Therefore, those subjects producing the greatest jump heights using the TOV+s method tended also to achieve the greatest jump heights when the other two methods were employed.

Taking TOV+s as the criterion measure, given the definition of jump height provided by Bobbert and van Ingen Schenau (1988), both TIA and TOV methods produce good correlation coefficients ($R > .890$) for both men and women (Tables 3 and 4). Although the differences were small for both men and women, the TIA method tended to produce smaller correlation coefficients than the TOV method. Using the correlation coefficient as a measure of validity for the different calculation methods, it appears that the TOV method displays greater validity as a method for calculating jump height when TOV+s is the criterion method. This may be expected given that the calculation for the TOV+s method uses TOV. However, validity scores between measures are affected by the reliability of the measures assessed (Ferguson & Takane, 1989). Recently, Moir (2008) investigated the intra-session stability reliability coefficients (ICC) for jump heights calculated from TOV+s, TIA, and TOV methods in men and women. TOV+s had the lowest reliability coefficients irrespective of gender, while TIA produced lower coefficients than TOV. Therefore, the lower validity coefficients reported here for the TIA method are in part due to the lower reliability of this method compared to TOV.

The estimation errors produced for the regression models were ≤ 0.025 m for both men and women, although again the errors associated with the TIA regression models were greater than those for the TOV models. These estimation errors are similar in magnitude to those reported by Aragón-Vargas (2000). It is worth noting that the slope of the regression model using the TIA jump height calculation method for men was different from 1.0 (coefficient of slope = 0.854; 95% CI = 0.732 – 0.976). This means that the error in the prediction of jump height using TIA will depend upon the magnitude of the calculated jump height, producing overestimations for low jump heights and slightly underestimating large jump heights. It is important to note also that this model produced the greatest prediction error (0.025 m).

Practical Recommendations

The different calculation methods used in the present study have logical validity, depending upon the definition of jump height used by the practitioner. Similarly, there was good consistency reported between the methods in both men and women. Therefore, perhaps the most important consideration when selecting a method of jump height calculation is the availability of equipment (e.g., contact mat, force platform, etc.). Through the use of a force platform, jump height defined as the vertical displacement achieved by the COM from take-off to the vertex of the flight trajectory (using TIA) or jump height defined in terms of take-off velocity (using TOV) and the COM position at take-off (using TOV+s) can be calculated. The TIA calculation method is the simplest to implement using force platform data, although the assumptions associated with the equation of uniform acceleration make this method prone to error. Through integration of the vertical force trace, the TOV method can be used. This method removes many of the confounding variables associated with TIA, although the practitioner must give careful consideration to the data collection variables (e.g., sampling frequency, estimation of body weight, etc.). Finally, through the double integration of the vertical force trace, Net COM can be calculated and then added to the jump height calculated from TOV. Although this is fairly straightforward to implement, the additional information gained from this procedure is limited. Based upon the findings of the present study, when using a force platform to calculate vertical jump height, practitioners are encouraged to use the TOV method.

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